

Stubby, Totally Stubby, & NSSA

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Purpose

The goal of this lab was to learn about and explore other aspects of the OSPF routing protocol, specifically stub areas, totally stubby areas, and NSSAs. Additionally, I had to figure out how to connect routers using different routing protocols to each other in a way that allows them to communicate with each other smoothly. In this lab, I connected routers using OSPF to a router using EIGRP in Cisco Packet Tracer.

Background Information

Introduction: OSPF & More

Before getting into the details about stub areas we must know - What is OSPF? What is it used for? When should you use it? In a nutshell, OSPF, which stands for Open Shortest Path First, is a routing protocol that uses LSAs or Link-State Advertisements and exchanges information between each router in its respective routing domain in order for all the routers to know what the topology of the whole network looks like. Some advantages of OSPF is that it supports various techniques which makes operation and exchanges between routers a lot easier, simpler, and smoother. Since the most efficient routes are being taken by packets (aka messages) and therefore involving less network traffic, this routing protocol is more economical than a protocol like RIP. Additionally, OSPF can be utilized as a hierarchical routing protocol, giving priority or a 'higher rank' to certain important parts of the network. This is extremely handy when dealing with larger and more complex networks.

How does this connect to the topic of this report? Well, OSPF networks can be split up into subdomains called areas, which are commonly identified by an area ID such as 'area 0'. Areas are similar to subnets that easily summarize routes and networks. Areas have benefits such as making a network more manageable and reducing the amount of routes that it needs to share. Areas do this by diluting and separating out the routes and then proceeding to give a brief run down on them.

Besides the broader grouping of 'areas' there are also Stub Areas, Not-So-Stubby Areas (also known as NSSA), and Totally Stub Areas. Each of these areas will be thoroughly discussed in this paper. First of all, some requirements of OSPF networks partitioned into different areas include having a backbone area, commonly defined in a topology as Area 0. A backbone area merges a set of individual areas into a single domain. Moreover, any area that is not a backbone area needs to be directly connected to the backbone. Finally, the backbone area cannot be split into smaller areas, such as during a system failure or when the link state is down.

When talking about the types of routers in a network running with OSPF, there is a Backbone router which includes at least a single interface in Area 0, an Internal Router who's interfaces are all in one area, an ABR or Area Border Router that includes at least one interface in a backbone area and a non-backbone area, and an ASBR, also known as an Autonomous System Border Router that has the ability to connect an area and an external AS or Autonomous System. Also, OSPF as I mentioned before, is a link-state based routing protocol, so we must discuss the different LSA or Link-State Advertisements of which there are seven main types.

Link-State Advertisement Types

LSA Type 1-Router: This type of LSA remains in the area it originates from and will also flood each router in the area. Type 1 LSAs show up in every area and it gives a description about devices that are directly connected to a router (this includes any virtual links) and is created by all internal routers in an area.

LSA Type 2-Network: Developed for the purpose of accommodating multi-access networks (a network with more than two machines), Type 2 LSAs include the information of every single route connected to the network the LSA originates from, the DR or Designated Router, and the prefix as well as the subnet mask. Like Type 1, this LSA is also found in all areas.

LSA Type 3-Summary: It has a misleading name as there is no summarization in OSPF. However, its function is to flood all areas in the OSPF network except the area of origin. This provides a network link ‘summary’ of the internal route and it is generated by an ABR.

LSA Type 4-Summary ASBR: Flooding all areas other than its own so that every OSPF router has the knowledge of the whereabouts of the ASBR. This flooding of areas is required since knowing where the ASBR lies should be known by all routers in OSPF. This LSA is established by the ABR.

LSA Type 5-Autonomous System External: Deals with External LSAs including external routes to the OSPF domain. This type of LSA, which is created by the ASBR, explains the routes which have been redistributed throughout the area.

LSA Type 6-Multicast OSPF: Not supported by Cisco (where instead Cisco used PIM aka Protocol Independent Multicast) or in other words, not used.

LSA Type 7-Not-So-Stubby Area: Has the same function and information as LSA Type 5, but since Type 5 is blocked by the NSSA, we use Type 7 instead. In other words, it is a replacement for Type 5 LSAs. LSA Type 7 is generated by an ASBR that is included in a NSSA.

Types of Areas

Normal/Standard Areas: These can be the backbone areas (where all areas connect to the central area), which is represented by Area 0. These areas are able to accept intra-area, inter-area, and external routes. In Normal Areas, default routes are not created by OSPF routers. What this area can do/know is, everything that is possible for an area of OSPF- there are no restrictions whatsoever. Each router knows about all the routers in the same area, however there is so much information being exchanged and shared at all times, that maintaining a full database is excessive and wasteful. Additionally, there are some routers that are incapable of handling large or full databases, specifically in big OSPF networks. Areas like these that cannot handle too much information. In a big network, we have the option of breaking it down into smaller areas, which is what we call Stub Areas.

Stub Areas: It can be called a ‘dead end’ within a network because the only way messages and updates can go in and out of this are through the ABR. This type of area allows inter and intra-area routes, but not advertisements from external routers, which further relieves OSPF routers with a database that is smaller and easier to handle. Since external advertisements are blocked, a default summary route is placed in the stub area so you have an alternate way of reaching the outlying routers and routes. If there are no external routes however, there is no need for defining or specifying if you have stub areas or not.

The purpose of a stub area is to make certain that routers will have the ability to route traffic over the network to external destinations without the hassle of dealing with every individual route that it does not need to be concerned with. The one restriction a stub area has is that there are no Type 5 LSAs allowed.

Not-So-Stubby Areas (NSSA): They are similar to stub areas but one difference between them is that NSSAs have more flexibility than stub areas, allowing them to import external routes into your OSPF routing domain and thus NSSAs can be used as a bridge between the OSPF routing domain and other smaller routing domains using a different routing technique. Also, an NSSA can serve as a stub or totally stubby area but with the added piece of an ASBR. NSSAs are useful when you are dealing with large networks for big companies or ISPs who require the ability to reach remote sites using an alternative protocol for routing. Similar to stub areas, NSSAs do not allow Type 5 LSAs. They stop AS-external LSAs (Type 5 LSAs) from flooding into any NSSAs. Due to this, it is important to put NSSAs at the edge of your OSPF routing domain. In addition, it still needs an LSA similar to it, which is where Type 7 LSA comes in to replace it.

(**Note:** When talking about a NSSA Totally Stubby Area, the restrictions are no Type 3,4,5 LSAs excluding the default summary route, but allows for Type 7 LSAs to replace Type 5 LSAs like in normal NSSAs)

Totally Stubby Areas: These can be called an extension to a stub area. Totally stub areas have routers keeping their LSDB-only information (Link-State Database) relating to routing inside their area. In other words, it does not advertise this information outside this area unless the routers use a default route. Default routes are placed into an area by an ABR, and with this route, the routers in the totally stubby areas will be able to use it to send traffic outside of their routing area. Even more than the other types of stub areas, this further reduces the amount of resources used by the routers in their area of the network that has no need for all the routing information in the OSPF routing domain. A totally stubby area restricts Type 3, 4, and 5 LSAs with the exception of default routes.

Definitions

ABR: An Area Border Router is a router that lies at the edge of one or more OSPF areas. It connects those areas to create a connection between the Backbone area and any other OSPF areas.

Areas: They are neighboring logical parts of a network that can be grouped together.

ASBR: An Autonomous System Boundary Router informs the OSPF routing domain of any external routes.

Backbone Area: The center of an OSPF network which lies in area 0. Every other area in the OSPF routing domain must be directly connected to the backbone area – this can include virtual links.

LSA: A Link-State Advertisement is a form of basic communication in OSPF for the IP or Internet Protocol. Within an LSA, there is information about a router's surrounding routing topology which is sent to other OSPF routers in the same area.

Normal/Standard Area: A typical area in an OSPF domain – an area that is not any kind of stub area. Standard areas include the Backbone area.

NSSA: A Not-So-Stubby Area is similar to a Stub area however it includes an ASBR.

Stub Area: These kinds of areas reduce the OSPF topology and make it easier for routers to process routing information since there is less to worry about.

Totally Stub Area: An area similar to a Stub area but restricts Type 3 LSAs from entering its area. Additionally, a Totally stubby area uses an ABR to inject the default route into an area.

Lab Summary

In this lab, I created eight networks and nine routers across four OSPF areas and one EIGRP area. These OSPF areas include the backbone area, stubby area, totally stubby area, and the NSSA. The stubby area, totally stubby area, NSSA, backbone area, and EIGRP area each had one router completely confined to their respective areas. Total, that is five routers. The remaining four routers are Area Border Routers (aka ABR). I put one between the backbone or standard area and the stubby area, one between the totally stubby area and the backbone, another between NSSA and the backbone, and finally an ABR between NSSA and EIGRP. I added one NIM-2T to all the routers, which were ISR4321 routers, with the exception of the middle router of the backbone area (or R1 in my topology) that had 2 NIM-2Ts added to it. To connect the routers to each other, I used serial DCE cables using interfaces, S0/1/0, S0/1/1, S0/2/1, and S0/2/0.

Lab Commands

Network *IP Address Wildcard Mask* area #: Identify which network and into what OSPF area the interface will go into

Redistribute EIGRP # Subnets: Imports routes from other routing protocols into EIGRP so routers using different protocols can communicate

Redistribute OSPF # Metric 1000 33 255 1 1500: Imports routes from other routing protocols into OSPF to allow routers using different protocols to communicate

Router-ID: Statically sets up the router's ID which can be used to identify the network device

Router OSPF #: Enables or starts the OSPF routing protocol on the router

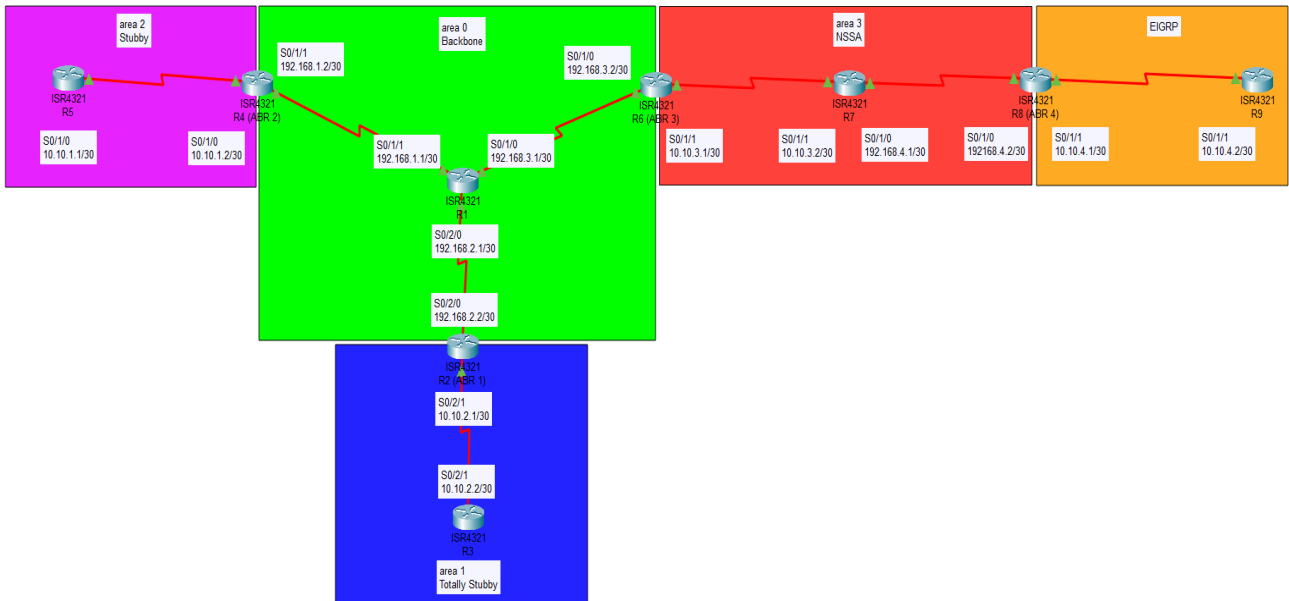
Show IP OSPF: Display information about one or more OSPF area

Show IP OSPF Database: Shows information related to different OSPF LSAs

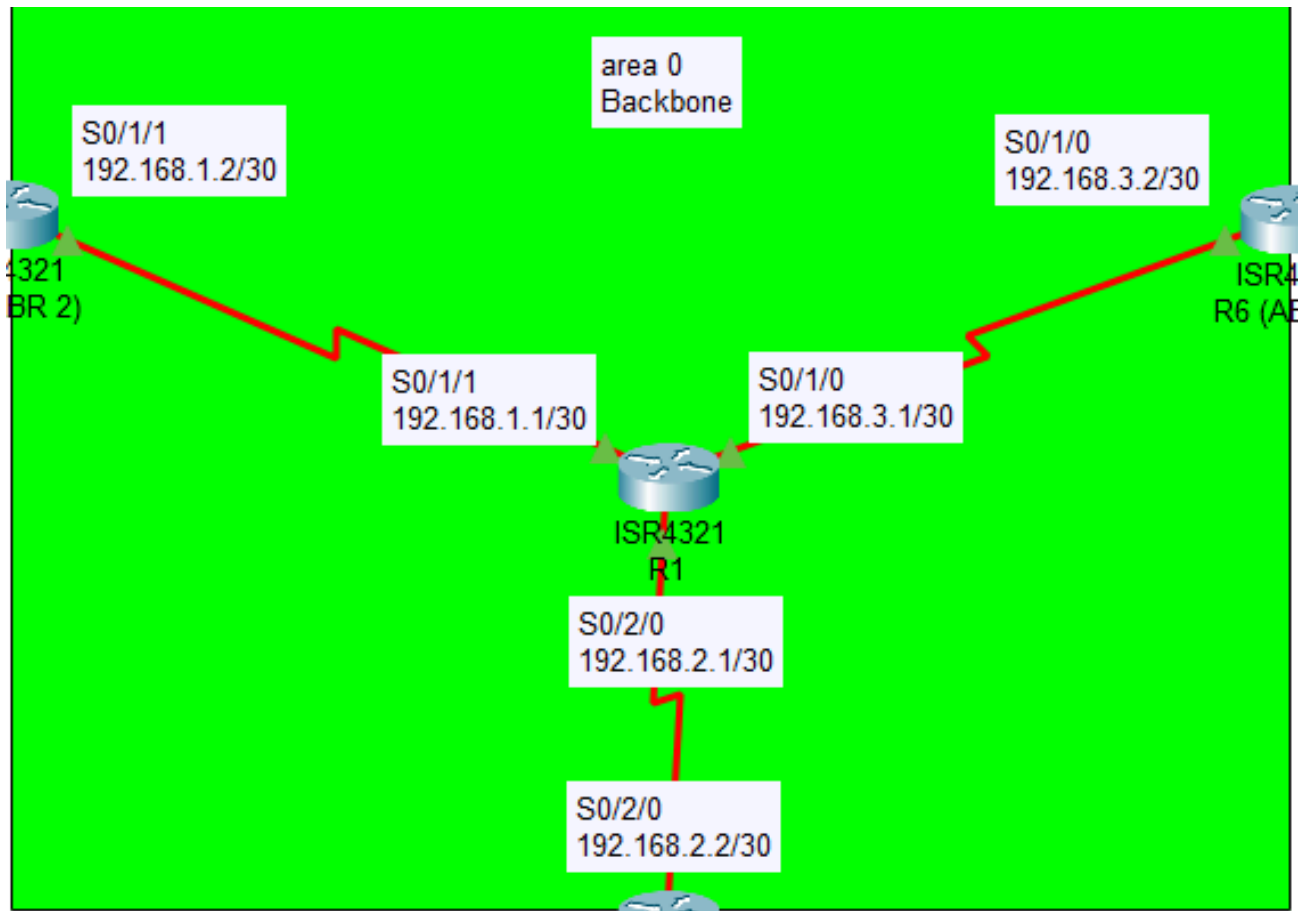
Show IP Route: Presents the routing table which has all the networks that the router can reach

Show Run: Shows all the configurations running on the RAM

Network Diagram

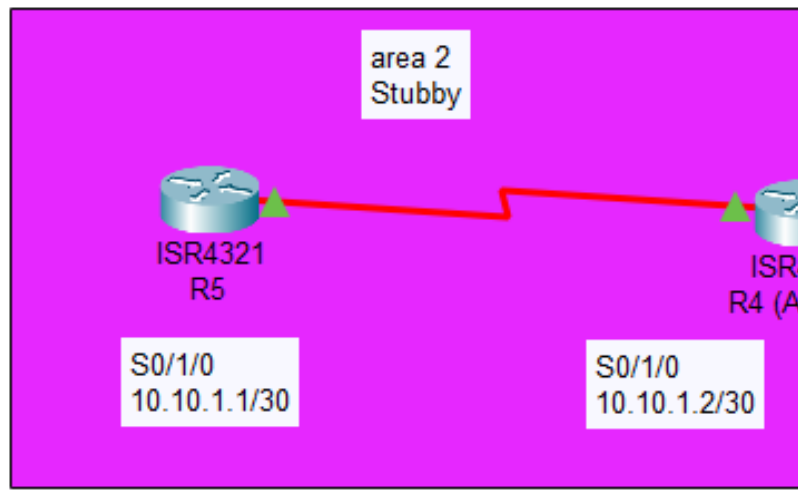


Area 0 Backbone:

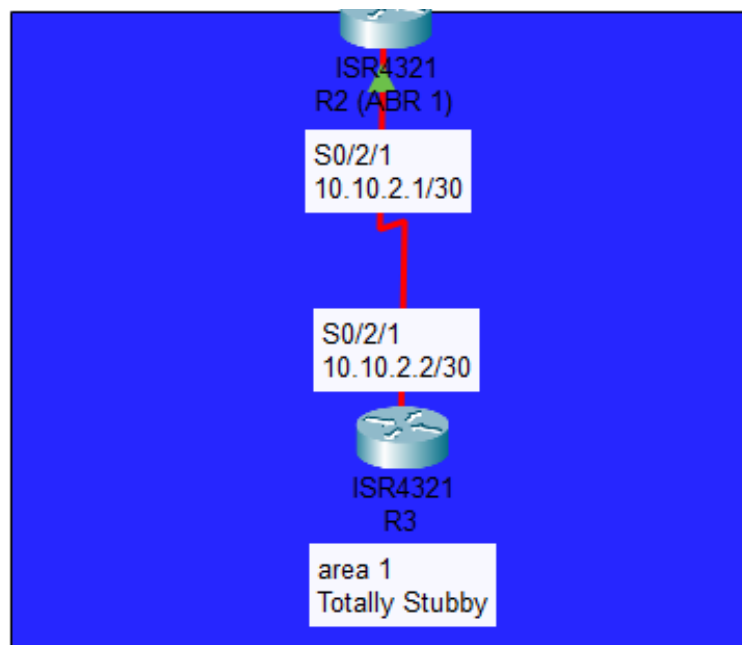


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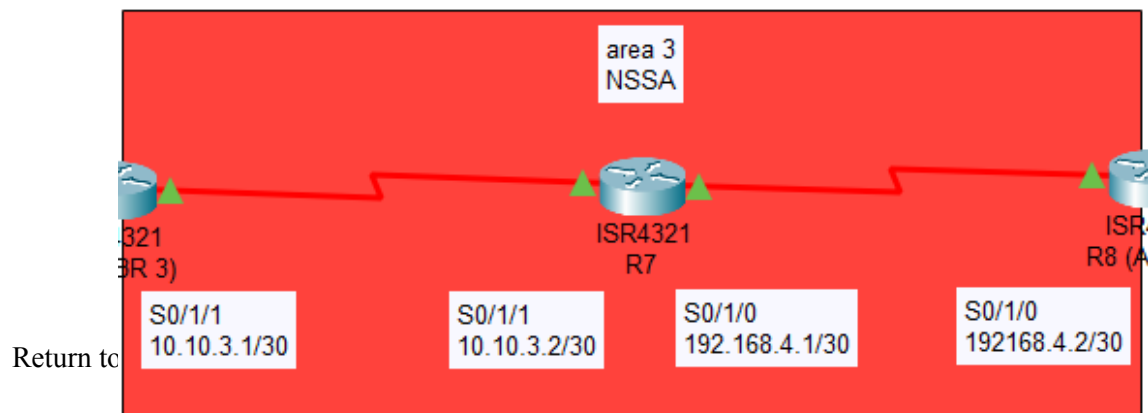
Area 1 Stubby:



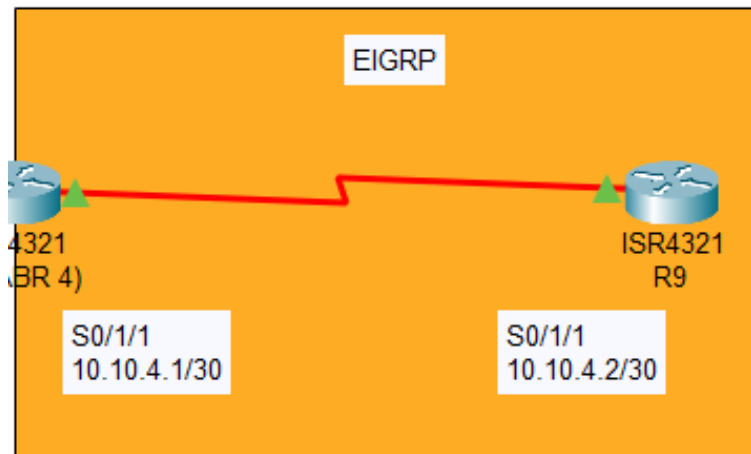
Area 2 Totally Stubby:



Area 3 NSSA:



EIGRP:



Configurations

Router 1:

- **Show Run**

```
interface Serial0/1/0
 ip address 192.168.3.1 255.255.255.252
 clock rate 2000000

interface Serial0/1/1
 ip address 192.168.1.1 255.255.255.252

interface Serial0/2/0
 ip address 192.168.2.1 255.255.255.252
 clock rate 2000000

router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 network 192.168.1.0 0.0.0.3 area 0
 network 192.168.2.0 0.0.0.3 area 0
 network 192.168.3.0 0.0.0.3 area 0
```

- **Show IP OSPF**

```
Routing Process "ospf 1" with ID 1.1.1.1
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10
 secs
```

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```

Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 1. Checksum Sum 0x006bef
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0
nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 3
    Area has no authentication
    SPF algorithm executed 10176 times
    Area ranges are
    Number of LSA 9. Checksum Sum 0x03cb89
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

- **Show IP OSPF Database**

```

OSPF Router with ID (1.1.1.1) (Process ID 1)

```

```

Router Link States (Area 0)

```

Link ID	ADV Router	Age	Seq#
Checksum Link count			
1.1.1.1	1.1.1.1	928	0x800000b4
0x0053cb 6			
4.4.4.4	4.4.4.4	930	0x8000005e
0x001b55 2			
2.2.2.2	2.2.2.2	929	0x8000005f
0x00d3a9 2			
6.6.6.6	6.6.6.6	928	0x8000005e
0x00cc8d 2			

```

Summary Net Link States (Area 0)

```

Link ID	ADV Router	Age	Seq#
Checksum			
10.10.3.0	6.6.6.6	1223	0x80004fe5
0x00714d			
10.10.2.0	2.2.2.2	1223	0x80000039
0x003c8f			
10.10.1.0	4.4.4.4	1209	0x80000039
0x000bb9			

```
192.168.4.0      6.6.6.6      621      0x80005062
0x003674
```

```
Summary ASB Link States (Area 0)
Link ID      ADV Router      Age      Seq#
Checksum
6.6.6.6      6.6.6.6      10      0x800050de
0x00bb33
```

```
Type-5 AS External Link States
Link ID      ADV Router      Age      Seq#
Checksum Tag
10.10.4.0    6.6.6.6      1223     0x80000039
0x006bef 0
```

- **Show IP Route**

Gateway of last resort is not set

```
10.0.0.0/30 is subnetted, 4 subnets
O IA    10.10.1.0/30 [110/128] via 192.168.1.2,
4294967277:4294967284:4294967246, Serial0/1/1
O IA    10.10.2.0/30 [110/128] via 192.168.2.2,
4294967277:4294967284:4294967246, Serial0/2/0
O IA    10.10.3.0/30 [110/128] via 192.168.3.2,
4294967277:4294967284:4294967246, Serial0/1/0
O E2    10.10.4.0/30 [110/20] via 192.168.3.2,
4294967277:4294967284:4294967246, Serial0/1/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.1.0/30 is directly connected, Serial0/1/1
L       192.168.1.1/32 is directly connected, Serial0/1/1
192.168.2.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.2.0/30 is directly connected, Serial0/2/0
L       192.168.2.1/32 is directly connected, Serial0/2/0
192.168.3.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.3.0/30 is directly connected, Serial0/1/0
L       192.168.3.1/32 is directly connected, Serial0/1/0
192.168.4.0/30 is subnetted, 1 subnets
O IA    192.168.4.0/30 [110/192] via 192.168.3.2,
4294967277:4294967284:4294967246, Serial0/1/0
```

Router 2 (ABR 1):

- **Show Run**

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```

interface Serial0/2/0
 ip address 192.168.2.2 255.255.255.252

interface Serial0/2/1
 ip address 10.10.2.1 255.255.255.252
 clock rate 2000000

router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
 area 1 stub no-summary
 network 10.10.2.0 0.0.0.3 area 1
 network 192.168.2.0 0.0.0.3 area 0

```

- **Show IP OSPF**

```

Routing Process "ospf 1" with ID 2.2.2.2
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 It is an area border router
 SPF schedule delay 5 secs, Hold time between two SPF's 10
secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 1. Checksum Sum 0x006bef
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 2. 1 normal 1 stub 0
nssa
 External flood list length 0
   Area 1
     Number of interfaces in this area is 1
     It is a stub area
       generates stub default route with cost 1
     Area has no authentication
     SPF algorithm executed 96 times
     Area ranges are
     Number of LSA 3. Checksum Sum 0x021840
     Number of opaque link LSA 0. Checksum Sum 0x000000
     Number of DCbitless LSA 0
     Number of indication LSA 0
     Number of DoNotAge LSA 0
     Flood list length 0
   Area BACKBONE(0)
     Number of interfaces in this area is 1
     Area has no authentication

```

```

SPF algorithm executed 10199 times
Area ranges are
Number of LSA 9. Checksum Sum 0x037eb0
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

```

- **Show IP OSPF Database**

OSPF Router with ID (2.2.2.2) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#
Checksum Link count			
2.2.2.2	2.2.2.2	1123	0x8000005f
0x00d3a9 2			
4.4.4.4	4.4.4.4	1125	0x8000005e
0x001b55 2			
1.1.1.1	1.1.1.1	1123	0x800000b4
0x0053cb 6			
6.6.6.6	6.6.6.6	1123	0x8000005e
0x00cc8d 2			

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#
Checksum			
10.10.2.0	2.2.2.2	1417	0x80000039
0x003c8f			
10.10.3.0	6.6.6.6	1418	0x80004fe5
0x00714d			
10.10.1.0	4.4.4.4	1404	0x80000039
0x000bb9			
192.168.4.0	6.6.6.6	816	0x80005062
0x003674			

Summary ASB Link States (Area 0)

Link ID	ADV Router	Age	Seq#
Checksum			
6.6.6.6	6.6.6.6	4	0x80005106
0x006a5c			

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#
Checksum Link count			
2.2.2.2	2.2.2.2	1123	0x80000060
0x00ca55 2			
3.3.3.3	3.3.3.3	1125	0x80000060
0x0067b4 2			

Summary Net Link States (Area 1)			
Link ID	ADV Router	Age	Seq#
Checksum			
0.0.0.0	2.2.2.2	1418	0x80000039
0x00e637			

Type-5 AS External Link States			
Link ID	ADV Router	Age	Seq#
Checksum Tag			
10.10.4.0	6.6.6.6	1418	0x80000039
0x006bef 0			

- **Show IP Route**

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O IA   10.10.1.0/30 [110/192] via 192.168.2.1,
4294967277:4294967287:4294967263, Serial0/2/0
C      10.10.2.0/30 is directly connected, Serial0/2/1
L      10.10.2.1/32 is directly connected, Serial0/2/1
O IA   10.10.3.0/30 [110/192] via 192.168.2.1,
4294967277:4294967287:4294967263, Serial0/2/0
O E2   10.10.4.0/30 [110/20] via 192.168.2.1,
4294967277:4294967287:4294967263, Serial0/2/0
    192.168.1.0/30 is subnetted, 1 subnets
O      192.168.1.0/30 [110/128] via 192.168.2.1,
4294967277:4294967287:4294967263, Serial0/2/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2
masks
C      192.168.2.0/30 is directly connected, Serial0/2/0
L      192.168.2.2/32 is directly connected, Serial0/2/0
    192.168.3.0/30 is subnetted, 1 subnets
O      192.168.3.0/30 [110/128] via 192.168.2.1,
4294967277:4294967287:4294967263, Serial0/2/0
    192.168.4.0/30 is subnetted, 1 subnets
O IA   192.168.4.0/30 [110/256] via 192.168.2.1,
4294967277:4294967287:4294967263, Serial0/2/0

```

Router 3:

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- **Show Run**

```
interface Serial0/2/1
 ip address 10.10.2.2 255.255.255.252

router ospf 1
 router-id 3.3.3.3
 log-adjacency-changes
 area 1 stub no-summary
 network 10.10.2.0 0.0.0.3 area 1
```

- **Show IP OSPF**

```
Routing Process "ospf 1" with ID 3.3.3.3
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10
 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 0 normal 1 stub 0
 nssa
 External flood list length 0
   Area 1
     Number of interfaces in this area is 1
     It is a stub area
     Area has no authentication
     SPF algorithm executed 136 times
     Area ranges are
     Number of LSA 3. Checksum Sum 0x021840
     Number of opaque link LSA 0. Checksum Sum 0x000000
     Number of DCbitless LSA 0
     Number of indication LSA 0
     Number of DoNotAge LSA 0
     Flood list length 0
```

- **Show IP OSPF Database**

```
OSPF Router with ID (3.3.3.3) (Process ID 1)
```

```
Router Link States (Area 1)
```

Link ID	ADV Router	Age	Seq#
Checksum	Link count		

3.3.3.3	3.3.3.3	1290	0x80000060
0x0067b4 2			
2.2.2.2	2.2.2.2	1289	0x80000060
0x00ca55 2			

Link ID	Summary Net	Link States (Area 1)	
Checksum	ADV Router	Age	Seq#
0.0.0.0	2.2.2.2	1584	0x80000039
0x00e637			

- **Show IP Route**

Gateway of last resort is 10.10.2.1 to network 0.0.0.0

```

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.2.0/30 is directly connected, Serial0/2/1
L       10.10.2.2/32 is directly connected, Serial0/2/1
O*IA 0.0.0.0/0 [110/65] via 10.10.2.1,
4294967277:4294967289:4294967289, Serial0/2/1

```

Router 4 (ABR 2):

- **Show Run**

```

interface Serial0/1/0
 ip address 10.10.1.2 255.255.255.252

interface Serial0/1/1
 ip address 192.168.1.2 255.255.255.252
 clock rate 2000000

router ospf 1
 router-id 4.4.4.4
 log-adjacency-changes
 area 2 stub
 network 10.10.1.0 0.0.0.3 area 2
 network 192.168.1.0 0.0.0.3 area 0

```

- **Show IP OSPF**

```

Routing Process "ospf 1" with ID 4.4.4.4
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 It is an area border router
 SPF schedule delay 5 secs, Hold time between two SPFs 10
 secs

```

```

Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 1. Checksum Sum 0x0069f0
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 2. 1 normal 1 stub 0
nssa
External flood list length 0
  Area 2
    Number of interfaces in this area is 1
    It is a stub area
      generates stub default route with cost 1
    Area has no authentication
    SPF algorithm executed 94 times
    Area ranges are
    Number of LSA 9. Checksum Sum 0x0527c7
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 1
    Area has no authentication
    SPF algorithm executed 10244 times
    Area ranges are
    Number of LSA 9. Checksum Sum 0x03b695
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

- **Show IP OSPF Database**

OSPF Router with ID (4.4.4.4) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#
Checksum Link count			
4.4.4.4	4.4.4.4	1668	0x8000005e
0x001b55 2			
2.2.2.2	2.2.2.2	1667	0x8000005f
0x00d3a9 2			
1.1.1.1	1.1.1.1	1667	0x800000b4
0x0053cb 6			

6.6.6.6 0x00cc8d 2	6.6.6.6	1667	0x8000005e
-----------------------	---------	------	------------

Summary Net Link States (Area 0)			
Link ID	ADV Router	Age	Seq#
Checksum			
10.10.1.0 0x0009ba	4.4.4.4	146	0x8000003a
192.168.4.0 0x003674	6.6.6.6	1359	0x80005062
10.10.3.0 0x0090be	6.6.6.6	160	0x80005153
10.10.2.0 0x003a90	2.2.2.2	160	0x8000003a

Summary ASB Link States (Area 0)			
Link ID	ADV Router	Age	Seq#
Checksum			
6.6.6.6 0x008fc9	6.6.6.6	9	0x80005173

Router Link States (Area 2)			
Link ID	ADV Router	Age	Seq#
Checksum Link count			
4.4.4.4 0x0062a6 2	4.4.4.4	1666	0x80000060
5.5.5.5 0x00e423 2	5.5.5.5	1670	0x8000005f

Summary Net Link States (Area 2)			
Link ID	ADV Router	Age	Seq#
Checksum			
0.0.0.0 0x00bce2	4.4.4.4	159	0x800001ae
192.168.1.0 0x006592	4.4.4.4	148	0x800001af
192.168.2.0 0x00e7d4	4.4.4.4	1652	0x800001a9
192.168.3.0 0x00dadf	4.4.4.4	1652	0x800001aa
192.168.4.0 0x00d166	4.4.4.4	1652	0x800001ab
10.10.3.0 0x0010bd	4.4.4.4	1652	0x800001ac
10.10.2.0 0x0019b4	4.4.4.4	1652	0x800001ad

Type-5 AS External Link States			
Link ID	ADV Router	Age	Seq#
Checksum Tag			
10.10.4.0	6.6.6.6	159	0x8000003a
0x0069f0 0			

- **Show IP Route**

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.10.1.0/30 is directly connected, Serial0/1/0
L       10.10.1.2/32 is directly connected, Serial0/1/0
O IA    10.10.2.0/30 [110/192] via 192.168.1.1,
4294967277:00:4294967253, Serial0/1/1
O IA    10.10.3.0/30 [110/192] via 192.168.1.1,
4294967277:00:4294967253, Serial0/1/1
O E2    10.10.4.0/30 [110/20] via 192.168.1.1,
4294967277:00:4294967253, Serial0/1/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.1.0/30 is directly connected, Serial0/1/1
L       192.168.1.2/32 is directly connected, Serial0/1/1
    192.168.2.0/30 is subnetted, 1 subnets
O       192.168.2.0/30 [110/128] via 192.168.1.1,
4294967277:00:4294967253, Serial0/1/1
    192.168.3.0/30 is subnetted, 1 subnets
O       192.168.3.0/30 [110/128] via 192.168.1.1,
4294967277:00:4294967253, Serial0/1/1
    192.168.4.0/30 is subnetted, 1 subnets
O IA    192.168.4.0/30 [110/256] via 192.168.1.1,
4294967277:00:4294967253, Serial0/1/1

```

Router 5:

- **Show Run**

```

interface Serial0/1/0
 ip address 10.10.1.1 255.255.255.252
 clock rate 2000000
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!

```

```

interface Vlan1
  no ip address
  shutdown
!
router ospf 1
  router-id 5.5.5.5
  log-adjacency-changes
  area 2 stub
  network 10.10.1.0 0.0.0.3 area 2

```

- **Show IP OSPF**

```

Routing Process "ospf 1" with ID 5.5.5.5
  Supports only single TOS(TOS0) routes
  Supports opaque LSA
  SPF schedule delay 5 secs, Hold time between two SPFs 10
secs
  Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
  Number of external LSA 0. Checksum Sum 0x000000
  Number of opaque AS LSA 0. Checksum Sum 0x000000
  Number of DCbitless external and opaque AS LSA 0
  Number of DoNotAge external and opaque AS LSA 0
  Number of areas in this router is 1. 0 normal 1 stub 0
nssa
  External flood list length 0
    Area 2
      Number of interfaces in this area is 1
      It is a stub area
      Area has no authentication
      SPF algorithm executed 265 times
      Area ranges are
      Number of LSA 9. Checksum Sum 0x0527c7
      Number of opaque link LSA 0. Checksum Sum 0x000000
      Number of DCbitless LSA 0
      Number of indication LSA 0
      Number of DoNotAge LSA 0
      Flood list length 0

```

- **Show IP OSPF Database**

```

OSPF Router with ID (5.5.5.5) (Process ID 1)

```

```

          Router Link States (Area 2)

```

Link ID	ADV Router	Age	Seq#
Checksum	Link	count	

5.5.5.5	5.5.5.5	1800	0x8000005f
0x00e423 2			
4.4.4.4	4.4.4.4	1797	0x80000060
0x0062a6 2			

Summary Net Link States (Area 2)			
Link ID	ADV Router	Age	Seq#
Checksum			
192.168.2.0	4.4.4.4	1783	0x800001a9
0x00e7d4			
192.168.3.0	4.4.4.4	1783	0x800001aa
0x00dadf			
192.168.4.0	4.4.4.4	1783	0x800001ab
0x00d166			
10.10.3.0	4.4.4.4	1783	0x800001ac
0x0010bd			
10.10.2.0	4.4.4.4	1783	0x800001ad
0x0019b4			
0.0.0.0	4.4.4.4	290	0x800001ae
0x00bce2			
192.168.1.0	4.4.4.4	279	0x800001af
0x006592			

- **Show IP Route**

Gateway of last resort is 10.10.1.2 to network 0.0.0.0

```

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.10.1.0/30 is directly connected, Serial0/1/0
L       10.10.1.1/32 is directly connected, Serial0/1/0
O IA    10.10.2.0/30 [110/256] via 10.10.1.2,
4294967278:4294967238:4294967254, Serial0/1/0
O IA    10.10.3.0/30 [110/256] via 10.10.1.2,
4294967278:4294967238:4294967254, Serial0/1/0
    192.168.1.0/30 is subnetted, 1 subnets
O IA    192.168.1.0/30 [110/128] via 10.10.1.2,
4294967278:4294967238:4294967254, Serial0/1/0
    192.168.2.0/30 is subnetted, 1 subnets
O IA    192.168.2.0/30 [110/192] via 10.10.1.2,
4294967278:4294967238:4294967254, Serial0/1/0
    192.168.3.0/30 is subnetted, 1 subnets
O IA    192.168.3.0/30 [110/192] via 10.10.1.2,
4294967278:4294967238:4294967254, Serial0/1/0
    192.168.4.0/30 is subnetted, 1 subnets
O IA    192.168.4.0/30 [110/320] via 10.10.1.2,
4294967278:4294967238:4294967254, Serial0/1/0

```

```
O*IA 0.0.0.0/0 [110/65] via 10.10.1.2,  
4294967278:4294967238:4294967254, Serial0/1/0
```

Router 6 (ABR 3):

- **Show Run**

```
interface Serial0/1/0  
 ip address 192.168.3.2 255.255.255.252  
  
interface Serial0/1/1  
 ip address 10.10.3.1 255.255.255.252  
 clock rate 2000000  
  
router ospf 1  
 router-id 6.6.6.6  
 log-adjacency-changes  
 area 3 nssa  
 network 10.10.3.0 0.0.0.3 area 3  
 network 192.168.3.0 0.0.0.3 area 0
```

- **Show IP OSPF**

```
Routing Process "ospf 1" with ID 6.6.6.6  
 Supports only single TOS(TOS0) routes  
 Supports opaque LSA  
 It is an area border router  
 SPF schedule delay 5 secs, Hold time between two SPFs 10  
secs  
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs  
 Number of external LSA 2. Checksum Sum 0x00d3e0  
 Number of opaque AS LSA 0. Checksum Sum 0x000000  
 Number of DCbitless external and opaque AS LSA 0  
 Number of DoNotAge external and opaque AS LSA 0  
 Number of areas in this router is 2. 1 normal 0 stub 1  
nssa  
 External flood list length 0  
 Area 3  
   Number of interfaces in this area is 1  
   It is a NSSA area  
   Perform type-7/type-5 LSA translation  
   Area has no authentication  
   SPF algorithm executed 148 times  
   Area ranges are  
   Number of LSA 9. Checksum Sum 0x04aed4  
   Number of opaque link LSA 0. Checksum Sum 0x000000  
   Number of DCbitless LSA 0
```

```

Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
Area BACKBONE(0)
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm executed 10323 times
Area ranges are
Number of LSA 9. Checksum Sum 0x0342cf
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

```

- **Show IP OSPF Database**

OSPF Router with ID (6.6.6.6) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#
Checksum Link count			
6.6.6.6	6.6.6.6	114	0x8000005f
0x00ca8e 2			
4.4.4.4	4.4.4.4	116	0x8000005f
0x001956 2			
2.2.2.2	2.2.2.2	115	0x80000060
0x00d1aa 2			
1.1.1.1	1.1.1.1	114	0x800000b5
0x0051cc 6			

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#
Checksum			
10.10.3.0	6.6.6.6	409	0x80005153
0x0090be			
192.168.4.0	6.6.6.6	1608	0x80005062
0x003674			
10.10.2.0	2.2.2.2	409	0x8000003a
0x003a90			
10.10.1.0	4.4.4.4	396	0x8000003a
0x0009ba			

Summary ASB Link States (Area 0)

Link ID	ADV Router	Age	Seq#
Checksum			

6.6.6.6	6.6.6.6	7	0x800051a7
0x0027fd			

Router Link States (Area 3)

Link ID	ADV Router	Age	Seq#
Checksum Link count			
6.6.6.6	6.6.6.6	116	0x80000060
0x004aa1 2			
8.8.8.8	8.8.8.8	117	0x80000061
0x001a14 2			
7.7.7.7	7.7.7.7	116	0x8000008b
0x002304 4			

Summary Net Link States (Area 3)

Link ID	ADV Router	Age	Seq#
Checksum			
192.168.3.0	6.6.6.6	393	0x8000012a
0x001d56			
192.168.2.0	6.6.6.6	1762	0x80000126
0x00b285			
192.168.1.0	6.6.6.6	1762	0x80000127
0x00bb7c			
10.10.2.0	6.6.6.6	1762	0x80000128
0x00e763			
10.10.1.0	6.6.6.6	1762	0x80000129
0x00f05a			

Type-7 AS External Link States (Area 3)

Link ID	ADV Router	Age	Seq#
Checksum Tag			
10.10.4.0	8.8.8.8	126	0x80000047
0x00c407 0			

Type-5 AS External Link States

Link ID	ADV Router	Age	Seq#
Checksum Tag			
10.10.4.0	6.6.6.6	407	0x8000003a
0x0069f0 0			

● Show IP Route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
 O IA 10.10.1.0/30 [110/192] via 192.168.3.1,
 4294967278:4294967240:4294967258, Serial0/1/0

```

O IA    10.10.2.0/30 [110/192] via 192.168.3.1,
4294967278:4294967240:4294967258, Serial0/1/0
C       10.10.3.0/30 is directly connected, Serial0/1/1
L       10.10.3.1/32 is directly connected, Serial0/1/1
O N2    10.10.4.0/30 [110/20] via 10.10.3.2,
4294967278:4294967240:4294967262, Serial0/1/1
        192.168.1.0/30 is subnetted, 1 subnets
O       192.168.1.0/30 [110/128] via 192.168.3.1,
4294967278:4294967240:4294967258, Serial0/1/0
        192.168.2.0/30 is subnetted, 1 subnets
O       192.168.2.0/30 [110/128] via 192.168.3.1,
4294967278:4294967240:4294967258, Serial0/1/0
        192.168.3.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.3.0/30 is directly connected, Serial0/1/0
L       192.168.3.2/32 is directly connected, Serial0/1/0
        192.168.4.0/30 is subnetted, 1 subnets
O       192.168.4.0/30 [110/128] via 10.10.3.2,
4294967278:4294967240:4294967262, Serial0/1/1

```

Router 7:

- **Show Run**

```

interface Serial0/1/0
 ip address 192.168.4.1 255.255.255.252
 clock rate 2000000

interface Serial0/1/1
 ip address 10.10.3.2 255.255.255.252

router ospf 1
 router-id 7.7.7.7
 log-adjacency-changes
 area 3 nssa
 network 10.10.3.0 0.0.0.3 area 3
 network 192.168.4.0 0.0.0.3 area 3

```

- **Show IP OSPF**

```

Routing Process "ospf 1" with ID 7.7.7.7
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10
secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000

```

```

Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 0 normal 0 stub 1
nssa
External flood list length 0
  Area 3
    Number of interfaces in this area is 2
    It is a NSSA area
    Perform type-7/type-5 LSA translation
    Area has no authentication
    SPF algorithm executed 247 times
    Area ranges are
    Number of LSA 9. Checksum Sum 0x0486e8
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

- **Show IP OSPF Database**

```

OSPF Router with ID (7.7.7.7) (Process ID 1)

```

```

Router Link States (Area 3)

```

Link ID	ADV Router	Age	Seq#
Checksum Link count			
7.7.7.7	7.7.7.7	255	0x8000008b
0x002304 4			
8.8.8.8	8.8.8.8	257	0x80000061
0x001a14 2			
6.6.6.6	6.6.6.6	256	0x80000060
0x004aa1 2			

```

Summary Net Link States (Area 3)

```

Link ID	ADV Router	Age	Seq#
Checksum			
192.168.3.0	6.6.6.6	533	0x8000012a
0x001d56			
192.168.2.0	6.6.6.6	101	0x8000012b
0x00a88a			
192.168.1.0	6.6.6.6	101	0x8000012c
0x00b181			
10.10.2.0	6.6.6.6	101	0x8000012d
0x00dd68			

```
10.10.1.0      6.6.6.6      101      0x8000012e
0x00e65f
```

```

Type-7 AS External Link States (Area 3)
Link ID      ADV Router      Age      Seq#
Checksum Tag
10.10.4.0    8.8.8.8      265     0x80000047
0x00c407 0
```

- **Show IP Route**

Gateway of last resort is not set

```

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O IA    10.10.1.0/30 [110/256] via 10.10.3.1,
4294967278:4294967242:4294967279, Serial0/1/1
O IA    10.10.2.0/30 [110/256] via 10.10.3.1,
4294967278:4294967242:4294967279, Serial0/1/1
C       10.10.3.0/30 is directly connected, Serial0/1/1
L       10.10.3.2/32 is directly connected, Serial0/1/1
O N2    10.10.4.0/30 [110/20] via 192.168.4.2,
4294967277:4294967247:4294967244, Serial0/1/0
192.168.1.0/30 is subnetted, 1 subnets
O IA    192.168.1.0/30 [110/192] via 10.10.3.1,
4294967278:4294967242:4294967279, Serial0/1/1
192.168.2.0/30 is subnetted, 1 subnets
O IA    192.168.2.0/30 [110/192] via 10.10.3.1,
4294967278:4294967242:4294967279, Serial0/1/1
192.168.3.0/30 is subnetted, 1 subnets
O IA    192.168.3.0/30 [110/128] via 10.10.3.1,
4294967278:4294967242:4294967279, Serial0/1/1
192.168.4.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.4.0/30 is directly connected, Serial0/1/0
L       192.168.4.1/32 is directly connected, Serial0/1/0
```

Router 8 (ABR 4):

- **Show Run**

```

interface Serial0/1/0
 ip address 192.168.4.2 255.255.255.252

interface Serial0/1/1
 ip address 10.10.4.1 255.255.255.252
 clock rate 2000000
```

```

router eigrp 1
  eigrp router-id 8.8.8.8
  redistribute ospf 1 metric 1000 33 255 1 1500
  network 10.10.4.0 0.0.0.3

```

```

router ospf 1
  router-id 8.8.8.8
  log-adjacency-changes
  area 3 nssa
  redistribute eigrp 1 subnets
  network 192.168.4.0 0.0.0.3 area 3

```

- **Show IP OSPF**

```

Router#show ip ospf
Routing Process "ospf 1" with ID 8.8.8.8
Supports only single TOS(TOS0) routes
Supports opaque LSA
It is an autonomous system boundary router
SPF schedule delay 5 secs, Hold time between two SPFs 10
secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 0 normal 0 stub 1
nssa
  External flood list length 0
    Area 3
      Number of interfaces in this area is 1
      It is a NSSA area
      Perform type-7/type-5 LSA translation
      Area has no authentication
      SPF algorithm executed 191 times
      Area ranges are
      Number of LSA 9. Checksum Sum 0x0486e8
      Number of opaque link LSA 0. Checksum Sum 0x000000
      Number of DCbitless LSA 0
      Number of indication LSA 0
      Number of DoNotAge LSA 0
      Flood list length 0

```

- **Show IP OSPF Database**

```

OSPF Router with ID (8.8.8.8) (Process ID 1)

```

Router Link States (Area 3)

Link ID	ADV Router	Age	Seq#
Checksum Link count			
8.8.8.8	8.8.8.8	384	0x80000061
0x001a14 2			
6.6.6.6	6.6.6.6	384	0x80000060
0x004aa1 2			
7.7.7.7	7.7.7.7	383	0x8000008b
0x002304 4			

Summary Net Link States (Area 3)

Link ID	ADV Router	Age	Seq#
Checksum			
192.168.3.0	6.6.6.6	661	0x8000012a
0x001d56			
192.168.2.0	6.6.6.6	229	0x8000012b
0x00a88a			
192.168.1.0	6.6.6.6	229	0x8000012c
0x00b181			
10.10.2.0	6.6.6.6	229	0x8000012d
0x00dd68			
10.10.1.0	6.6.6.6	229	0x8000012e
0x00e65f			

Type-7 AS External Link States (Area 3)

Link ID	ADV Router	Age	Seq#
Checksum Tag			
10.10.4.0	8.8.8.8	393	0x80000047
0x00c407 0			

● Show IP Route

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O IA   10.10.1.0/30 [110/320] via 192.168.4.1,
4294967278:4294967244:4294967285, Serial0/1/0
O IA   10.10.2.0/30 [110/320] via 192.168.4.1,
4294967278:4294967244:4294967285, Serial0/1/0
O      10.10.3.0/30 [110/128] via 192.168.4.1,
4294967278:4294967244:4294967285, Serial0/1/0
C      10.10.4.0/30 is directly connected, Serial0/1/1
L      10.10.4.1/32 is directly connected, Serial0/1/1
    192.168.1.0/30 is subnetted, 1 subnets
O IA   192.168.1.0/30 [110/256] via 192.168.4.1,
4294967278:4294967244:4294967285, Serial0/1/0

```

```

    192.168.2.0/30 is subnetted, 1 subnets
O IA    192.168.2.0/30 [110/256] via 192.168.4.1,
4294967278:4294967244:4294967285, Serial0/1/0
    192.168.3.0/30 is subnetted, 1 subnets
O IA    192.168.3.0/30 [110/192] via 192.168.4.1,
4294967278:4294967244:4294967285, Serial0/1/0
    192.168.4.0/24 is variably subnetted, 2 subnets, 2
masks
C        192.168.4.0/30 is directly connected, Serial0/1/0
L        192.168.4.2/32 is directly connected, Serial0/1/0

```

Router 9:

- **Show Run**

```

interface Serial0/1/0
  no ip address
  clock rate 2000000
  shutdown

interface Serial0/1/1
  ip address 10.10.4.2 255.255.255.252

router eigrp 1
  eigrp router-id 9.9.9.9
  redistribute ospf 1 metric 1000 33 255 1 1500
  network 10.10.4.0 0.0.0.3

```

- **Show IP Route**

```

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D EX    10.10.1.0/30 [170/3080448] via 10.10.4.1,
4294967278:4294967248:4294967287, Serial0/1/1
D EX    10.10.2.0/30 [170/3080448] via 10.10.4.1,
4294967278:4294967248:4294967287, Serial0/1/1
D EX    10.10.3.0/30 [170/3080448] via 10.10.4.1,
4294967278:4294967248:4294967287, Serial0/1/1
C        10.10.4.0/30 is directly connected, Serial0/1/1
L        10.10.4.2/32 is directly connected, Serial0/1/1
    192.168.1.0/30 is subnetted, 1 subnets
D EX    192.168.1.0/30 [170/3080448] via 10.10.4.1,
4294967278:4294967248:4294967287, Serial0/1/1
    192.168.2.0/30 is subnetted, 1 subnets
D EX    192.168.2.0/30 [170/3080448] via 10.10.4.1,
4294967278:4294967248:4294967287, Serial0/1/1

```

192.168.3.0/30 is subnetted, 1 subnets
D EX 192.168.3.0/30 [170/3080448] via 10.10.4.1,
4294967278:4294967248:4294967287, Serial0/1/1
192.168.4.0/30 is subnetted, 1 subnets
D EX 192.168.4.0/30 [170/3080448] via 10.10.4.1,
4294967278:4294967249:4294967242, Serial0/1/1

LSA Packets

Stub Area Router (R5) and ABR between Stub and Backbone Area (R4 aka ABR 2)

OSPF LSA Header

0 16 24 Bits		
LSAAGE:0	OPTIONS:0	LSA TYPE:3
LINK STATE ID:192.168.1.0		
ADVERTISING ROUTER:4.4.4.4		
LS SEQUENCE NUM:-2147483646		
LS CHECKSUM:49891	LENGTH:28	

3.835	--	R1	OSPF
3.836	R6 (ABR 3)	R1	OSPF
3.836	R5	R4 (ABR 2)	OSPF
3.836	R1	R4 (ABR 2)	OSPF
3.839	--	R2 (ABR 1)	OSPF
3.840	R2 (ABR 1)	R1	OSPF

Reset Simulation ☒ Constant Delay Captured to: 3.840 s

Play Controls

NSSA Router (R7) and ABR between NSSA and EIGRP Area (R8 aka ABR 4)

OSPF LSA Header

0 16 24 Bits		
LSAAGE:9	OPTIONS:10	LSA TYPE:7
LINK STATE ID:10.10.4.0		
ADVERTISING ROUTER:8.8.8.8		
LS SEQUENCE NUM:-2147483647		
LS CHECKSUM:20928	LENGTH:36	

22.182	R6 (ABR 3)	R7	OSPF
22.182	R7	R6 (ABR 3)	OSPF
22.210	--	R7	OSPF
22.210	--	R8 (ABR 4)	OSPF
22.211	--	R3	OSPF
22.211	--	R5	OSPF
22.211	R7	R8 (ABR 4)	OSPF
22.211	R8 (ABR 4)	R7	OSPF
22.211	--	R1	OSPF
22.212	R3	R2 (ABR 1)	OSPF
22.212	R5	R4 (ABR 2)	OSPF

Totally Stubby Router (R3) and ABR between Totally Stubby and Backbone Area (R2 aka ABR 1)

OSPF LSA Header

0 16 24 Bits		
LSAAGE:0	OPTIONS:0	LSA TYPE:3
LINK STATE ID:0.0.0.0		
ADVERTISING ROUTER:2.2.2.2		
LS SEQUENCE NUM:-2147483647		
LS CHECKSUM:22526	LENGTH:28	

27.216	--	R6 (ABR 3)	OSPF
27.216	--	R5	OSPF
27.216	--	R1	OSPF
27.216	R6 (ABR 3)	R1	OSPF
27.216	--	R2 (ABR 1)	OSPF
27.217	R3	R2 (ABR 1)	OSPF
27.217	R7	R6 (ABR 3)	OSPF
27.217	R6 (ABR 3)	R7	OSPF
27.217	R5	R4 (ABR 2)	OSPF
27.217	R1	R4 (ABR 2)	OSPF
27.217	R2 (ABR 1)	R1	OSPF

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Problems

One issue I faced was when I began setting up multi-area OSPF and the one EIGRP area after configuring the IP addresses and subnet masks. As I was writing the network statements in OSPF, I didn't check what I had written in each route. When I was done, I tried to make sure everything could ping to confirm that the routing protocols were working. Nothing appeared to work. I went back and did a "show run" on all the routers and realized I wrote a few of the network statements incorrectly, such as writing the wrong area or messing up the IP address. Both when trying to initially write the network statements for routing and when I tried fixing it, I got confused. I decided I should label my network topology on Packet Tracer and color code the different areas so I would not be confused. After that, I was able to successfully input all the network statements correctly and get multi-area OSPF to work.

Another issue I had was when I finished configuring everything—OSPF, EIGRP, specifying stubby, totally stubby, and NSSA, connecting EIGRP to OSPF. I could not ping a router from one area to another area. I looked over my configurations again and I had other people check my work and everything seemed to look correct. Finally, one friend suggested I wait a day with Packet Tracer open and when I checked on my network the next day, everything was running and able to ping.

Lastly, I was trying to find certain LSAs in packets between routers in the stub and backbone area, totally stubby and backbone area, and between NSSA and EIGRP. I was able to find the LSA I was looking for in the stub area (a type 3 between R4 [aka ABR 2] and R5 for the stub area), but could not find the other two LSAs I was searching for no matter how long I let the simulation packets run, despite reopening Packet Tracer, or power cycling the devices. I would check every packet to see if they had the LSAs I was looking for and I double, and triple checked my configurations alongside my friends and finally we found I had not set router-ids on my EIGRP routers. At first, we thought that router-id wasn't that important, so we tried looking for any other differences. But when we discovered no others, I decided to give it a shot and add the router-id. I did a "copy run start" command and then power cycled the devices one more time and lo and behold, I was quickly able to find the packets I needed (a type 7 LSA between R7 and R8 [aka ABR 4] for NSSA to EIGRP and a type 3 between R2 [aka ABR 1] and R3 for the totally stubby area).

Conclusion

By setting up nine routers in OSPF stubby areas, totally stubby areas, and NSSAs as well as connecting the multi-area OSPF to an EIGRP area was a good introduction to configuring a larger network. Previously, I had only really worked with five routers max, but this lab used nine which allowed me to learn how to prepare and organize my network setup. I had to make a clear topology and label each router, network, area, and interface so that I would not get confused. Additionally, this lab gave me the chance to learn more detail about the OSPF routing protocol and how to set up specific areas on my own, what those areas are used for, what their restrictions are, and more. With this lab, I had to learn about and try to understand stub areas mostly on my own and figure out how to get it running. The result of this lab was that I was able to add to my knowledge of networking and learn that I am capable of teaching myself something new if necessary so that I can build a better network.

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